Worksheet 2A: Lines, Coordinates & Vectors

Part A: Coordinate Basics (10 problems)

1. Plot the following points on a coordinate plane and label each with its coordinates: - A = (3, 4) - B = (-2, 5) - C = (-4, -3) - D = (5, -2) - E = (0, 3) - F = (-3, 0)

2. In which quadrant (I, II, III, IV) or on which axis does each point lie? - (a) (7, -2) - (b) (-3, -8) - (c) (0, 5) - (d) (-4, 6) - (e) (2, 0)

3. Find the distance between each pair of points: - (a) (0, 0) and (3, 4) - (b) (1, 2) and (4, 6) - (c) (-2, 3) and (4, -1) - (d) (-5, -2) and (3, 4)

4. Find the midpoint of the segment connecting: - (a) (2, 8) and (6, 2) - (b) (-3, 4) and (5, -2) - (c) (0, 7) and (-4, -1)

5. A point starts at (2, 3) and moves 4 units right and 5 units down. What are its new coordinates?

6. The midpoint of segment PQ is (3, 1). If P = (7, 4), what are the coordinates of Q?

Part B: Vectors (12 problems)

Vectors are written in bold: **v**, **w**, **u**. They have both magnitude (length) and direction.

7. Given $\mathbf{v} = (3, 4)$ and $\mathbf{w} = (1, -2)$, calculate: - (a) $\mathbf{v} + \mathbf{w}$ - (b) $\mathbf{v} - \mathbf{w}$ - (c) $\mathbf{w} - \mathbf{v}$ - (d) $2\mathbf{v}$ - (e) $-3\mathbf{w}$ - (f) $2\mathbf{v} + 3\mathbf{w}$

8. Find the magnitude (length) of each vector. Recall: $|\mathbf{v}| = \sqrt{(v_1^2 + v_2^2)}$ - (a) $\mathbf{v} = (3, 4)$ - (b) $\mathbf{w} = (5, 12)$ - (c) $\mathbf{u} = (-2, 2)$ - (d) $\mathbf{p} = (1, 1)$

9. A unit vector has magnitude 1. Find the unit vector in the same direction as: - (a) $\mathbf{v} = (3, 4)$ - (b) $\mathbf{w} = (0, 5)$

(Hint: divide each component by the magnitude)

10. Vector $\mathbf{a} = (4, 3)$ and vector $\mathbf{b} = (-3, 4)$. - (a) Calculate $\mathbf{a} + \mathbf{b}$ - (b) Calculate $|\mathbf{a}|$ and $|\mathbf{b}|$ - (c) Calculate $|\mathbf{a} + \mathbf{b}|$ - (d) Is $|\mathbf{a} + \mathbf{b}|$ equal to $|\mathbf{a}| + |\mathbf{b}|$? Why or why not?

11. A drone is at position (2, 5). It moves by vector $\mathbf{v} = (3, -2)$, then by vector $\mathbf{w} = (-1, 4)$. What is its final position?

12. The zero vector is $\mathbf{0} = (0, 0)$. For any vector **v**: - (a) What is $\mathbf{v} + \mathbf{0}$? - (b) What is $\mathbf{v} + (-\mathbf{v})$? - (c) What vector **w** satisfies $\mathbf{v} + \mathbf{w} = \mathbf{0}$?

13. Given points P = (1, 2) and Q = (4, 6), find the vector $\mathbf{PQ}$ that goes from P to Q.

(Hint: $\boldsymbol{PQ}$ = Q - P)

14. If **v** = (2, 5) and **w** = (k, 15) are parallel (same direction), what is k?

Part C: Slope and Linear Equations (14 problems)

15. Calculate the slope between each pair of points: - (a) (0, 0) and (4, 8) - (b) (1, 3) and (5, 11) - (c) (2, 7) and (6, 3) - (d) (-3, 2) and (4, 2) - (e) (5, 1) and (5, 9)

16. Without calculating, identify the slope as positive, negative, zero, or undefined: - (a) A line going uphill from left to right - (b) A line going downhill from left to right - (c) A horizontal line - (d) A vertical line

17. Convert to slope-intercept form (y = mx + b): - (a) 2x + y = 7 - (b) 3x - 2y = 12 - (c) x + 4y = 8 - (d) 5x - y + 3 = 0

18. Find the equation of the line with: - (a) Slope 3 and y-intercept -2 - (b) Slope -½ and y-intercept 4 - (c) Slope 0 and y-intercept 5

19. Find the equation of the line passing through: - (a) (0, 3) with slope 2 - (b) (2, 5) with slope -1 - (c) (4, -1) with slope ¾ - (d) (-2, 6) with slope 0

20. Find the equation of the line passing through both points: - (a) (1, 2) and (3, 8) - (b) (0, 5) and (4, 1) - (c) (-2, -3) and (4, 9)

21. Find the x-intercept and y-intercept of each line: - (a) y = 2x - 6 - (b) 3x + 4y = 12 - (c) y = -½x + 3

22. Line A has equation y = 3x + 1. Line B has equation y = 3x - 4. - (a) What are their slopes? - (b) Are they parallel, perpendicular, or neither? - (c) Will they ever intersect?

23. Line A has slope 2. What is the slope of a line perpendicular to A?

24. Find the equation of the line: - (a) Parallel to y = 2x + 1 and passing through (3, 5) - (b) Perpendicular to y = 3x - 2 and passing through (6, 1)

Part D: Systems of Linear Equations (12 problems)

Solve each system by substitution or elimination. State if there's no solution or infinitely many.

25.

$$y = 2x + 1$$

$$y = -x + 7$$

26.

$$x + y = 10$$

$$x - y = 2$$

27.

$$2p + 3q = 13$$

$$p - q = 1$$

28.

$$3m + 2n = 12$$

$$6m + 4n = 24$$

29.

$$2r + 5s = 11$$

$$4r + 10s = 15$$

30.

$$\frac{t}{2} + u = 5$$

$$t - 2u = 4$$

31. Three coffees and two teas cost €11. Two coffees and three teas cost €12. Find the price of each.

32. A rectangle's perimeter is 28 cm. Its length is 4 cm more than its width. Find the dimensions.

Graphical Interpretation:

33. Without solving algebraically, sketch these lines and estimate their intersection point:

$$y = x + 1$$

$$y = -2x + 7$$

34. What does it mean graphically when a system has: - (a) Exactly one solution? - (b) No solution? - (c) Infinitely many solutions?

35. The lines y = 2x + 3 and y = 2x - 1 form a system. Without graphing or substituting, explain why there's no solution.

36. Write a system of two equations that has: - (a) The solution (2, 5) - (b) No solution - (c) Infinitely many solutions

---